Guidelines for Camera Matching

Thatcher Freeman

github.com/thatcherfreeman

Abstract

Two cameras can be approximately matched by a 3x3 matrix, applied in the scene linear domain. The selection of this matrix is done by minimizing the error between the patches on a color chart shot with both cameras, under lighting representative of the intended shooting conditions. This document will explain how to best conduct the photography needed to perform this matching process.

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I. Prerequisites

The camera match is generally dependent on the choice of lens, the lighting illuminant, codec, white balance setting, and sometimes the base ISO and exposure used. Select these such that they are representative of the intended shooting conditions. The more that the actual setup differs from the profiling setup, the less accurate the camera match will be.

It is also necessary to record with a log encoding which we can easily linearized (eg, supported by DaVinci Resolve or ACES or via a custom linearization profile). We have separate documentation for if your camera is not supported by these systems, and if you're unsure, feel free to reach out and we can help you profile the camera's log curve.

The above must be selected and used for each camera intended to be matched.

II. Color Chart

We recommend using a Calibrite ColorChecker Classic color chart (pictured in Figure 1). If using a different color chart, then select one that has a wide range of non-grey colors, including some skin tones. The ColorChecker Video chart is **NOT** recommended due to its low color diversity.



Figure 1: *Composition and Chart Angle Guide*

III. Procedure

Follow the below requirements with each camera to be matched. Each camera should be independently white balanced and their positions (and particularly their angle with respect to the chart) should be approximately matched. The quality of the color match is dependent on the degree to which this procedure is followed and to how similar these conditions are to the real production camera configuration and lighting.

i. Requirements

1. **Angle the chart to control glare** - The light should be placed at such an angle to the color chart that the camera is not in-line with the primary reflection of the light in the color chart. IE if the color chart were a mirror, the light's reflection should not be visible to the camera. If the camera is normal to the test chart, putting the light at a 45 degree angle to the test chart would easily avoid this. See Figure 1
2. **Even lighting on the color chart** - Place the light far from the test chart so that the light hitting the chart is uniform (place the light at least 4x the length of the long side of the test chart away).
3. **Correct white balance** - The camera should be white balanced for this light source. If white balancing via Kelvins is to be used, choose the best Kelvin white balance for the light, for each camera. Alternatively, AWB off of a white card illuminated by your light source. If the camera is to be used with a single white balance for the entire production, just use that white balance.
4. **Proper exposure** - Properly expose the color chart with a light meter. Ensure that no chip on the color chart is clipped, and make sure that it's not underexposed.
5. **Color chart fills the middle third of the image** - This is a compromise between maximizing the number of pixels we can sample for each color chip, and avoiding lens vignetting. Fill the middle rule of thirds box with the color chart.
6. **Dark background material to control lens flaring** - We don't want there to be any bright spots in the background of the image that could result in lens flaring, as this would contaminate the chart colors.
7. **Lens stopped down a couple stops** - By stopping down the lens, contrast and flare are typically improved. Pick an aperture in the range of f/4-f/8, if possible.
8. **Lock off the tripod** - By keeping the chart and tripod stationary, this makes it easier for us to denoise the chart and extract its true colors
9. **1-5s Duration** - It is only necessary to shoot the chart for between 1 and 5 seconds.
10. **Do not obscure the color chart** - Do not let shadows or fingers or clamps cover any part of the color chips on the chart.

IV. Edge Cases

i. Rapidly Changing Lighting Conditions

If the test has to be shot under rapidly variable lighting conditions, then it is advisable to put your cameras next to each other and record the chart simultaneously.

ii. Colored Filtration

If using filters that have a strong color cast, it may be useful to build the camera match for a single camera under each filter in question so we can match them to the hero setup. This is simply a matter of following the procedure with one camera but for each filter or lens.